

Study of models of cosmic inflation and observational constraints on their parameters

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Contents

1	Introduction	2
2	General relativity tools for cosmology	3
2.1	Equivalence principle	3
2.2	Review of tensors, covariant and contravariant vectors	3
2.3	Christoffel symbols	4
2.4	Geodesic equation	5
2.5	Covariant derivative:	5
2.6	Parallel transport	6
2.7	Riemann tensor:	7
2.8	Ricci tensor and scalar:	7
2.9	Einstein-Hilbert action:	8
2.10	Einstein equation:	9
2.11	FLRW metric:	10
2.12	Friedmann equations	10
3	Inflation theory	13
3.1	The horizon problem	13
3.2	The flatness problem	13
3.3	Inflation and Scalar Field Evolution	13
3.4	The Slow-roll approximation	15
3.5	Cosmological Signatures of Inflation	17
3.6	e-folds for quadratic potential	18
3.7	Inflation models depending on n_s and r parameters	19
4	Conclusion	21
	Bibliographie	22

Chapter 1

Introduction

The theory of cosmic inflation, developed in the late 1970s and early 1980s, was introduced to explain the present structure of our Universe and several observations of the cosmic microwave background (CMB). Since the period before the CMB cannot be directly observed with current technology, our description of it is mainly based on theoretical extrapolations. The homogeneity and isotropy of the CMB observed by the COBE, WMAP and Planck satellites [1, 2] gives us important information about the evolution of our universe.

Inflation concerns a very short period of time after the Big Bang, around 10^{-36} s to 10^{-32} s depending on models. During this very short period called the very early Universe, our universe would have expanded by a factor of approximately 10^{30} . This theory tries to explain why opposite regions of the CMB, which should not have been in causal contact, have almost the same temperature. In this sense, inflation gives an explanation for the observed homogeneity and isotropy of the CMB.

This rapid expansion can be described by a theoretical scalar field, called the inflaton. At the end of inflation, the inflaton field oscillates around the minimum of its potential and decays into radiation. This ends the inflaton-dominated period and marks the beginning of reheating.

In this internship, we will first introduce the basic tools needed to understand the geometry of the Universe within the framework of General Relativity. We will then apply Einstein's equations to the FLRW metric in order to study cosmology. In a second part, we will focus on inflation theory, its motivations and its main parameters. Finally, we will study reheating during chaotic inflation and follow the evolution of matter and radiation densities in the very early Universe.

Chapter 2

General relativity tools for cosmology

For all the following statements we will assume that the speed of light in vacuum is $c = 1$, and choose the cosmological metric convention $(-, +, +, +)$. We will also assume the Einstein summation convention.

2.1 Equivalence principle

The equivalence principle is a fundamental observation made by Einstein, which allow us to distinguish gravitation from other forces by defining gravity as an inertial effect. Locally, the acceleration produced by a gravitational field in an inertial frame is indistinguishable from the acceleration of a non-inertial frame. A freely falling frame, neglecting tidal forces, is also indistinguishable from an inertial frame.

Since special relativity is only valid in inertial frames, it follows that it is a local approximation of general relativity (GR). General relativity extends the study to non-inertial frames through the curvature of spacetime, and predicts the motion of particles in spacetime along paths called geodesics.

2.2 Review of tensors, covariant and contravariant vectors

To study curved spaces we need a metric tensor $g_{\mu\nu}$, which is a structure attached to a manifold allowing distances and angles to be defined. Tensors have the property to transform consistently under changes of coordinates. A tensor $T(k, l)$ is indeed a mathematical object describing multi-linear connections from a collection of k dual vectors T_p^* (linear maps) and l vectors T_p to $\mathbb{R}$ so:

$$T : \underbrace{T_p^* \times T_p^* \times \dots \times T_p^*}_{k \text{ times}} \times \underbrace{T_p \times T_p \times \dots \times T_p}_{l \text{ times}} \Rightarrow \mathbb{R}$$

where " $\times$ " denotes the Cartesian product.

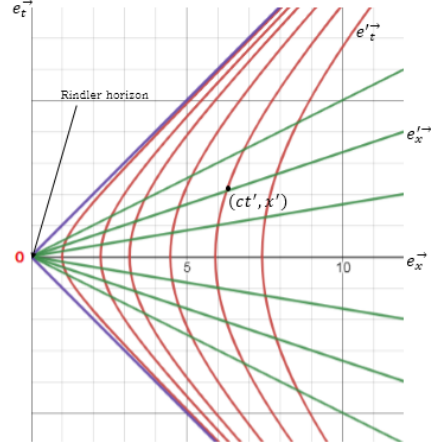
In this way, we can describe some common types of tensors: a $(0, 0)$ tensor corresponds to

a scalar, a $(1,0)$ tensor to a vector, a $(0,1)$ tensor to a linear form, and a $(1,1)$ tensor to a matrix.

To study general relativity, it is also essential to physically understand the notion of covariant and contravariant vectors. A great way to understand these notions while staying in a special relativity framework is by studying the construct of Rindler coordinates.

The usual description of an event by Minkowski coordinates (ct, x) is no longer adapted to accelerated frames. Rindler coordinates distort the time and space axes depending on the position of the event (ct', x') . They describe the point of view of a uniformly accelerated observer in flat spacetime, whose trajectory is hyperbolic. A contravariant vector v^i , for the velocity vector $\vec{V} = v^i e_i$, follows the coordinate lines telling how coordinates change. Whereas the covariant vector v_i would stick to the surface of constant coordinates so $v_i = \vec{V} \cdot e_i$. The coordinates being related to the metric we can deduce: $x_i = g_{ij}x^j$ and $x^i = g^{ij}x_j$. We understand there that bases and vectors change depending on the metric in accelerated frames, which is fundamental in general relativity.

The Rindler coordinates grid with the $\vec{e}_t, \vec{e}_x$ time and space axes. An event (ct', x') shows a change of axes in $\vec{e}_t, \vec{e}_x$. *Illustration adapted from online educational material.*



2.3 Christoffel symbols

For the study of the geodesics of particles, we extend the Minkowski metric $diag(-1, +1, +1, +1)$, which describes spacetime in the absence of gravitation, to curved metrics. To measure distances and compare vectors at different points, we introduce the Christoffel symbols. They are coefficients that describe the connection between tangent spaces at neighbouring points, or in other words the Levi-Civita connection. [3, 4]

Christoffel symbols tell us how the relation between covariant and contravariant vectors varies in a space with a specific metric $g_{\mu\nu}$. They describe how the basis vectors change

from point to point:

$$\Gamma_{\mu\nu}^{\rho} = \frac{1}{2}g^{\rho\sigma}(\partial_{\mu}g_{\sigma\nu} + \partial_{\nu}g_{\mu\sigma} - \partial_{\sigma}g_{\mu\nu}) \quad (2.1)$$

2.4 Geodesic equation

The equivalence principle tells us that without any external forces, the path taken by a particle satisfies the geodesic equation. This means that the tangential component of the acceleration vanishes. The particle follows the straightest possible path in curved spacetime, influenced only by the geometry. We need to describe the curves where the acceleration is normal to the spacetime surface, so only curvature will influence the particle: Taking a 4-vector $\vec{R}(u, v)$, in spacetime, with the vectors $\vec{e}_u \equiv \frac{\partial R}{\partial u}$, $\vec{e}_v \equiv \frac{\partial R}{\partial v}$ and deriving with an affine parameter λ , two times, to get acceleration we have:

$$\begin{aligned} \frac{d^2 R}{d\lambda^2} &= \frac{d}{d\lambda} \left(\frac{\partial R}{\partial U} \frac{dU}{d\lambda} + \frac{\partial R}{\partial V} \frac{dV}{d\lambda} \right) \\ &= \frac{d^2 U}{d\lambda^2} \frac{\partial R}{\partial U} + \frac{d^2 V}{d\lambda^2} \frac{\partial R}{\partial V} + \left(\frac{dV}{d\lambda} \right)^2 \frac{\partial^2 R}{\partial V^2} + \left(\frac{dU}{d\lambda} \right)^2 \frac{\partial^2 R}{\partial U^2} + 2 \frac{dV}{d\lambda} \frac{dU}{d\lambda} \frac{\partial^2 R}{\partial U \partial V} \\ &= \frac{d^2 U^i}{d\lambda^2} \frac{\partial R}{\partial U^i} + \frac{dU^i}{d\lambda} \frac{dU^j}{d\lambda} \frac{\partial^2 R}{\partial U^i \partial U^j} \end{aligned} \quad (2.2)$$

The tangential accelerations being null, we can decompose the last term that embed tangential and normal acceleration:

$$\frac{\partial^2 R}{\partial U^i \partial U^j} = \Gamma_{ij}^k \frac{\partial R}{\partial U^k} + L_{ij} n \quad (2.3)$$

where Γ_{ij}^k are the Christoffel symbols, which encode the metric properties of the tangent space, and $L_{ij}n$ is the normal component of the acceleration, which does not have to vanish in curved space.

To only keep the normal acceleration from $\frac{d^2 x^k}{d\lambda^2}$, we can compute the following relation that should be null, corresponding to the Geodesic equation path:

$$0 = \frac{d^2 x^k}{d\lambda^2} + \Gamma_{ij}^k \frac{dx^i}{d\lambda} \frac{dx^j}{d\lambda} \quad (2.4)$$

Or the corresponding equation of the Geodesic curve:

$$0 = \frac{d^2 R}{d\lambda^2} - L_{ij} \frac{dx^i}{d\lambda} \frac{dx^j}{d\lambda} \quad (2.5)$$

2.5 Covariant derivative:

As we saw before, derivatives in curved space have to take into account the change of basis from point to point. The covariant derivative is defined in order to preserve tensorial transformation properties. First, we can see that an ordinary derivative does not preserve

these properties under a coordinate transformation $x^\mu \rightarrow x'^\mu$, because an extra term appears:

$$U'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} U^\nu \quad (2.6)$$

$$\partial_\alpha U'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} \partial_\alpha U^\nu + \underbrace{\partial_\alpha \left(\frac{\partial x'^\mu}{\partial x^\nu} \right)}_{\text{parasite term}} U^\nu \quad (2.7)$$

We therefore define a new derivative, called the covariant derivative, which includes the Christoffel symbols:

$$\nabla_\alpha U^\mu = \partial_\alpha U^\mu + \Gamma_{\alpha\beta}^\mu U^\beta \quad (2.8)$$

For a covariant vector we use :

$$\nabla_\alpha (U^\mu U_\mu) = \partial_\alpha (U^\mu U_\mu), \quad (2.9)$$

This impose

$$\nabla_\alpha U_\mu = \partial_\alpha U_\mu - \Gamma_{\alpha\mu}^\beta U_\beta \quad (2.10)$$

2.6 Parallel transport

Parallel transport describes how a vector moves along a curve in spacetime while remaining geometrically “parallel”. In flat space, a vector can be transported without changing its direction or magnitude, but in curved spacetime these quantities may depend on the path taken.

For a vector V^μ transported along a curve $x^\mu(\lambda)$ parametrized by λ , the condition of parallel transport is

$$\frac{DV^\mu}{D\lambda} = 0 \quad (2.11)$$

where $\frac{D}{D\lambda}$ is the covariant derivative along the trajectory:

$$\frac{DV^\mu}{D\lambda} = \frac{dV^\mu}{d\lambda} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\lambda} V^\beta \quad (2.12)$$

As a consequence, even if a vector is parallel transported, its components may change depending on the geometry of spacetime.

2.7 Riemann tensor:

As we already know, spacetime can be described as a 4-dimensional pseudo-Riemannian manifold with a Lorentzian metric. This metric defines an inner product on each tangent space of the manifold, varying smoothly from point to point, which is necessary to study curved spaces intrinsically.

The Riemann tensor gives us information about the holonomy of vectors transported in a curved space. It can be defined as a failure of covariant derivatives to commute. In other words, it shows that the direction and magnitude of vectors may vary depending on the path followed in curved space, which characterizes curvature. By considering the non-commutation of two different parallel transports of a vector we can calculate it. As an example, the variation of $\vec{T}$ through a loop composed of $\vec{V}$ and $\vec{W}$, described by a factor $R(\vec{V}, \vec{W})\vec{T}$, is:

$$R(\vec{V}, \vec{W})\vec{T} = V^\mu W^\nu T^\sigma R(e_\mu, e_\nu)e_\sigma \quad (2.13)$$

The Riemann tensor is given by:

$$R(\vec{e}_\mu, \vec{e}_\nu)\vec{e}_\sigma = \nabla_{\vec{e}_\mu} \nabla_{\vec{e}_\nu} \vec{e}_\sigma - \nabla_{\vec{e}_\nu} \nabla_{\vec{e}_\mu} \vec{e}_\sigma - \nabla_{[\vec{e}_\mu, \vec{e}_\nu]} \vec{e}_\sigma \quad (2.14)$$

Where $[\vec{e}_\mu, \vec{e}_\nu] = 0$ in coordinate bases, using the covariant derivative we have:

$$R(\vec{e}_\mu, \vec{e}_\nu)\vec{e}_\sigma = \nabla_{\vec{e}_\mu} (\Gamma_{\sigma\nu}^\rho \vec{e}_\sigma) - \nabla_{\vec{e}_\nu} (\Gamma_{\mu\sigma}^\rho \vec{e}_\sigma) \quad (2.15)$$

And so the final expression:

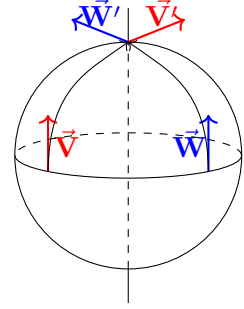
$$R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma_{\nu\sigma}^\rho - \partial_\nu \Gamma_{\mu\sigma}^\rho + \Gamma_{\mu\lambda}^\rho \Gamma_{\nu\sigma}^\lambda - \Gamma_{\nu\lambda}^\rho \Gamma_{\mu\sigma}^\lambda \quad (2.16)$$

We can use the metric to lower the up index:

$$R_{\rho\sigma\mu\nu} = g_{\rho\lambda} R^\lambda_{\sigma\mu\nu} \quad (2.17)$$

2.8 Ricci tensor and scalar:

The Ricci tensor $R_{\mu\nu}$ is obtained by contracting the Riemann tensor, reducing a tensor with 256 components to one with 16 components. It provides information about how spacetime curvature affects volumes locally. The Riemann tensor can be contracted in several ways, but due to its symmetries only one independent contraction gives the Ricci tensor :



Extrinsic view of the non-commutation of vectors $\vec{V}$ and $\vec{W}$, transported along geodesics to become $\vec{V}'$ and $\vec{W}'$ in a spherical space.

$$R^\rho_{\mu\rho\nu} = -R^\rho_{\mu\nu\rho} \equiv R_{\mu\nu} \quad (2.18)$$

A further contraction gives us a scalar quantity, because corresponding to the trace of a 4×4 matrix. This gives us a simple indication of the curvature of spacetime. We contract the Ricci tensor and obtain the Ricci scalar:

$$R = R^\alpha_\alpha = g^{\mu\nu} R_{\mu\nu} \quad (2.19)$$

2.9 Einstein-Hilbert action:

The Einstein-Hilbert action defines the Einstein field equations. Its gravitational part is written as:

$$S = C \int d^4x R \sqrt{-g} \quad (2.20)$$

where R is the Ricci scalar, $R = g^{\mu\nu} R_{\mu\nu}$, and $g = \det(g_{\mu\nu})$ is the determinant of the metric tensor. The constant C is physically determined as $\frac{1}{16\pi G}$.

We can develop the Einstein-Hilbert action with infinitesimal variation:

$$\delta S = C \int d^4x (\sqrt{-g} \delta R + R \delta \sqrt{-g}) = \int d^4x \sqrt{-g} [\delta R + R \frac{\delta \sqrt{-g}}{\sqrt{-g}}] \quad (2.21)$$

Using logarithm properties and matrix trace properties, we can find that:

$$R \frac{\delta \sqrt{-g}}{\sqrt{-g}} = \frac{1}{2} R \delta \log(-g) = \frac{1}{2} R \delta \log(\det(A)) = \frac{1}{2} R \delta \text{Tr}[\delta \log(A)] \quad (2.22)$$

and using $\frac{\delta \sqrt{-g}}{\sqrt{-g}} = g^{-1} \delta g$ we find:

$$= \frac{1}{2} R \delta \text{Tr}[g^{-1} \delta g] = -\frac{1}{2} R g^{\mu\nu} \delta g_{\mu\nu}$$

Using

$$\delta R = (\delta g^{\mu\nu}) R_{\mu\nu} + g^{\mu\nu} \delta R_{\mu\nu} \quad (2.23)$$

Where $g^{\mu\nu} \delta R_{\mu\nu}$ is a total divergence term, so does not contribute to the equations of motion after integration. Using :

$$\delta \sqrt{-g} = -\frac{1}{2} \sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu}, \quad (2.24)$$

we obtain

$$\delta S = C \int d^4x \sqrt{-g} \left(R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \right) \delta g^{\mu\nu} \quad (2.25)$$

Since the action must be stationary for variations $\delta g^{\mu\nu}$,

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 0 \quad (2.26)$$

which defines the very famous Einstein tensor:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \quad (2.27)$$

2.10 Einstein equation:

The Einstein tensor introduced above :

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} \quad (2.28)$$

gives us a tensor quantity that describes the curvature of spacetime due to gravitation, independently of coordinate changes. A fundamental property of the Einstein tensor is that its covariant divergence is null:

$$\nabla^\mu G_{\mu\nu} = 0 \quad (2.29)$$

as a consequence of the contracted Bianchi identities $\nabla_\rho R^\rho_\mu = \frac{1}{2}\nabla_\mu R$.

We now have to introduce the stress tensor, or energy-momentum tensor, which describes the distribution and flow of energy and momentum in spacetime. Generally written as :

$$T_{\mu\nu} \quad (2.30)$$

It acts as the source term in Einstein's field equations and contains different components such as:

$$\underbrace{T_{00}}_{\text{Energy density}} \quad \underbrace{T_{0i}}_{\text{Momentum density and energy flux}} \quad \underbrace{T_{ij}}_{\text{Pressure and mechanical stresses}} \quad (2.31)$$

This is essential since the stress tensor also satisfies a local conservation law:

$$\nabla^\mu T_{\mu\nu} = 0 \quad (2.32)$$

Einstein postulated that spacetime curvature must be proportional to the energy-momentum tensor content, so that :

$$G_{\mu\nu} = \kappa T_{\mu\nu} \quad (2.33)$$

where κ is a proportionality constant.

By requiring the weak-field limit of General Relativity to reproduce Newtonian gravity, and comparing with the Poisson equation :

$$\nabla^2 \phi = 4\pi G\rho \quad (2.34)$$

The Einstein field equations are

$$G_{\mu\nu} = 8\pi GT_{\mu\nu} \quad (2.35)$$

Giving the non-vacuum Einstein equation:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi GT_{\mu\nu} \quad (2.36)$$

Adding a cosmological constant we have:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu} \quad (2.37)$$

Because of the smoothness of our Universe on large scales, larger than galaxy clusters, cosmologists make the perfect fluid approximation

$$T_{\mu\nu} = (\rho + P)u_\mu u_\nu + P g_{\mu\nu} \quad (2.38)$$

where ρ is the energy density, P the pressure, and u^μ the four-velocity of the fluid.

2.11 FLRW metric:

The gravitation is only the manifestation of the curved geometry of our universe: the matter and radiation in our universe being curving spacetime. We can study our universe with curved spaces, as spherical, flat or hyperbolic, by creating a metric that will allow us to embed the characteristics we observed around us. The metric we will develop, called the Robertson-Walker metric, is the most general way to describe an expanding homogeneous and isotropic universe :

We can express the Friedmann-Lemaitre-Robertson-Walker space time interval:

$$ds^2 = -dt^2 + a(t)^2 \left[\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right] \quad (2.39)$$

With $a(t)$ the scale factor, where $k = 1$ for a spherical geometry, $k = -1$ for hyperbolic geometry, and $k = 0$ for flat space. We can write the FLRW metric as:

$$g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & \frac{a(t)^2}{1 - Kr^2} & 0 & 0 \\ 0 & 0 & a(t)^2 r^2 & 0 \\ 0 & 0 & 0 & a(t)^2 r^2 \sin^2 \theta \end{pmatrix} \quad (2.40)$$

2.12 Friedmann equations

We can see the Friedmann equations as an adaptation of Einstein's field equations to the FLRW metric, which are otherwise too general to be directly applied to cosmological observations. They allow us to go from tensor equations with many parameters to two differential equations that describe the expansion and composition of the Universe. To find them, we write the metric in tensor notation :

$$ds^2 = -dt^2 + a^2(t) \gamma_{ij} dx^i dx^j \quad (2.41)$$

with γ_{ij} the spatial components of the metric. The non-zero components of the metric and its inverse are :

$$g_{tt} = -1, \quad g_{ij} = a^2(t) \gamma_{ij} \quad (2.42)$$

$$g^{tt} = -1, \quad g^{ij} = \frac{1}{a^2(t)} \gamma^{ij} \quad (2.43)$$

Differentiating these time-dependent elements, we find:

$$\partial_t g_{ij} = 2a\dot{a} \gamma_{ij} \quad (2.44)$$

and

$$\partial_k g_{ij} = a^2 \partial_k \gamma_{ij} \quad (2.45)$$

We include them into the Christoffel symbols of the metric :

$$\Gamma_{ij}^t = a\dot{a} \gamma_{ij} \quad (2.46)$$

$$\Gamma_{tj}^i = \Gamma_{jt}^i = \frac{\dot{a}}{a} \delta_j^i \quad (2.47)$$

Using them, we can compute the corresponding Ricci tensor, whose non-vanishing components are:

$$R_{00} = -3\frac{\ddot{a}}{a} \quad (2.48)$$

$$R_{ij} = (a\ddot{a} + 2\dot{a}^2 + 2k) \gamma_{ij} \quad (2.49)$$

where k is the spatial curvature parameter.

The Ricci scalar is then obtained from the contraction

$$R = g^{\mu\nu} R_{\mu\nu},$$

which gives

$$R = 6 \left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2} + \frac{k}{a^2} \right) \quad (2.50)$$

Using the definition of the Einstein tensor

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \quad (2.51)$$

we obtain the non-vanishing components

$$G_{00} = 3 \left(\frac{\dot{a}^2}{a^2} + \frac{k}{a^2} \right) \quad (2.52)$$

$$G_{ij} = - \left(2\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2} + \frac{k}{a^2} \right) g_{ij} \quad (2.53)$$

By introducing the assumption of perfect fluid stress tensor and the Einstein field equation :

$$T_{\mu\nu} = (\rho + P)u_\mu u_\nu + P g_{\mu\nu}, \quad G_{\mu\nu} = 8\pi G T_{\mu\nu} \quad (2.54)$$

We get the two Friedmann equations:

$$\textbf{First equation} : \left\{ \left(\frac{\dot{a}}{a} \right)^2 = \frac{8\pi G}{3} \rho - \frac{k}{a^2} + \frac{\Lambda}{3} \right. \quad (2.55)$$

$$\textbf{Second equation} : \left\{ \frac{\ddot{a}}{a} = -\frac{4\pi G}{3} (\rho + 3P) + \frac{\Lambda}{3} \right. \quad (2.56)$$

The Friedmann equations [1, 2] govern the dynamical evolution of our Universe with the scale factor parameter $a(t)$ and the curvature component k . The first Friedmann equation relates the expansion rate of the Universe to its energy density ρ . The second equation determines the acceleration of the expansion where both density and pressure contribute to the dynamics. A positive pressure increases gravitational attraction slowing the acceleration, while negative pressure contributes to accelerated expansion. We can observe that the Λ dark energy term is the one responsible for the acceleration of the expansion, an accelerated expansion means :

$$\rho + 3P < 0 \quad (2.57)$$

This implies $\ddot{a} > 0$. For a decelerated expansion we have :

$$\rho + 3P > 0 \quad (2.58)$$

This implies $\ddot{a} < 0$. This way we can argue that the cosmological acceleration condition is :

$$P < -\frac{\rho}{3} \quad (2.59)$$

Chapter 3

Inflation theory

3.1 The horizon problem

Despite the success of the Big Bang theory, we should remember that our description of the very early Universe is still based on extrapolations. Our observations are limited to the CMB, and the inflation of the early universe before recombination remains inaccessible to current technologies.

The horizon problem arises from the incredible homogeneity of the CMB. Regions separated by more than two degrees on the sky should not have been in causal contact in the standard Big Bang model without inflation. For example, two points of opposite directions in the CMB have very similar temperature and density of matter, but we know that without inflation they have never communicated despite their similarities. Indeed, concerning the temperature fluctuations we observe that:

$$\frac{\Delta T}{T} \approx 10^{-5}$$

This temperature homogeneity is called the horizon problem.

3.2 The flatness problem

$$\Omega - 1 = \frac{k}{a^2 H^2} \tag{3.1}$$

For the very early universe we observe that $|\Omega - 1| \ll 10^{-60}$

3.3 Inflation and Scalar Field Evolution

A common approach to describe inflation is to introduce inflatons, particles responsible for inflation in a scalar field denoted by ϕ in an expanding spacetime. In a homogeneous and isotropic universe, the scalar field is only time dependent, we have the relation $\phi = \phi(t)$.

The interaction between the field and gravity occurs with the metric tensor, without explicit coupling between ϕ and curvature. We can write the total action as :

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} - \frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - U(\phi) \right] \quad (3.2)$$

where $U(\phi)$ is the scalar field potential and R is the Ricci scalar. In GR, the stress tensor is obtained by varying the matter part of the action so:

For the scalar field Lagrangian,

$$\mathcal{L}_\phi = -\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - U(\phi) \quad (3.3)$$

this variation give

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S_{\text{matter}}}{\delta g^{\mu\nu}} \quad (3.4)$$

$$T_{\mu\nu} = \partial_\mu \phi \partial_\nu \phi + g_{\mu\nu} \mathcal{L}_\phi \quad (3.5)$$

Because of the perfect fluid assumption we can find the corresponding energy density and pressure for this classical field :

$$\rho_\phi = \frac{\dot{\phi}^2}{2} + U(\phi) \quad (3.6)$$

$$P_\phi = \frac{\dot{\phi}^2}{2} - U(\phi) \quad (3.7)$$

To use this relations we can assume a flat FLRW geometry and have :

$$ds^2 = -dt^2 + a^2(t) (dx^2 + dy^2 + dz^2) \quad (3.8)$$

Reducing Einstein equations to :

$$H^2 = \frac{8\pi G}{3} \left(\frac{\dot{\phi}^2}{2} + U(\phi) \right) \quad (3.9)$$

$$\frac{\ddot{a}}{a} = -\frac{8\pi G}{3} \left(\frac{\dot{\phi}^2}{2} - U(\phi) \right) \quad (3.10)$$

where $H = \frac{\dot{a}}{a}$ is the Hubble expansion rate.

The field equation obtained from the action is

$$\ddot{\phi} + 3H\dot{\phi} + \frac{dU}{d\phi} = 0 \quad (3.11)$$

The first term is the one responsible for the acceleration of our field motion. The second show an damping effect of the cosmic expansion on the scalar field evolution, where the field dynamics is damped by the Hubble expansion.

We can assume this way that an inflationary period occur when the scale factor accelerates :

$$\ddot{a} > 0 \quad \rho_\phi + 3P_\phi < 0 \quad (3.12)$$

But the potential energy has to dominates over the kinetic contribution so the accelerated expansion condition becomes :

$$\dot{\phi}^2 \ll U(\phi) \quad (3.13)$$

In this limit, with the equations of pressures and density for classical field, we can do the approximation that :

$$P_\phi \simeq -\rho_\phi \quad (3.14)$$

which approximate to a constant expansion rate and an exponential growth of the scale factor :

$$a(t) \sim e^{Ht} \quad (3.15)$$

Which shows accelerated expansion with negative pressure when the scalar field dominates the kinetic energy, this situation is called the slow-roll approximation.

3.4 The Slow-roll approximation

In most inflationary models the scalar field evolves slowly compared to the expansion rate of the Universe [1, 5, 6]. This regime is called the slow-roll approximation where $\dot{\phi}^2 \ll U(\phi)$ while the acceleration of the field is low compared to the Hubble friction term :

$$|\ddot{\phi}| \ll |3H\dot{\phi}| \quad (3.16)$$

Under these conditions, the Friedmann equation simplifies so :

$$H^2 \simeq \frac{8\pi G}{3} U(\phi) \quad (3.17)$$

showing that the scalar field potential is the dominant parameter in the expansion rate.

We can reduce

$$\ddot{\Phi}(t) + 3H\dot{\Phi}(t) + \frac{dU}{d\phi} = 0 \quad (3.18)$$

with this approximation so :

$$3H\dot{\phi} + \frac{dU}{d\phi} \simeq 0 \quad (3.19)$$

The slow evolution of the scalar field allows inflation to last long enough to solve the horizon and flatness problems discussed previously. The Hubble parameter varies slowly during this period, giving the quasi-exponential expansion form of the scale factor we had before.

To validate the slow-roll regime, we need to introduce some observable slow-roll parameters. Starting from the definition :

$$\epsilon \equiv -\frac{\dot{H}}{H^2} \quad (3.20)$$

We use the slow-roll Friedmann equation :

$$H^2 \simeq \frac{U(\phi)}{3M_{\text{Pl}}^2} \quad (3.21)$$

where the reduced Planck mass is $M_{\text{Pl}}^2 = \frac{1}{8\pi G}$.

And we differentiate to find that :

$$3H\dot{\phi} \simeq -U'(\phi) \quad (3.22)$$

so

$$\dot{H} \simeq -\frac{U'(\phi)^2}{18M_{\text{Pl}}^2 H^2} \quad (3.23)$$

Using again $H^2 \simeq U/(3M_{\text{Pl}}^2)$, we have the **first parameter** :

$$\epsilon = \frac{M_{\text{Pl}}^2}{2} \left(\frac{U'(\phi)}{U(\phi)} \right)^2 \quad (3.24)$$

Same way defining :

$$\eta \equiv -\frac{\ddot{\phi}}{H\dot{\phi}} \quad (3.25)$$

and differentiating $3H\dot{\phi} \simeq -U'$, we have :

$$\eta \simeq \frac{U''(\phi)}{3H^2} \quad (3.26)$$

The **second parameter** :

$$\eta = M_{\text{Pl}}^2 \frac{U''(\phi)}{U(\phi)} \quad (3.27)$$

These quantities allow to characterize the validity of the slow-roll approximation directly from the shape of the inflaton potential. They permit to measure how slowly the field evolves compared to the Hubble expansion rate. Inflation will occur only when both of these parameters are small, $\epsilon \ll 1$ and $|\eta| \ll 1$, permitting the potential energy to dominates. In these conditions we can observe that our dynamics is practically in a De Sitter regime.

The end of inflation is generally defined by the condition

$$\epsilon \simeq 1 \quad (3.28)$$

corresponding to the point where the accelerated expansion stops and the scalar field begins to oscillate around the minimum of its potential. We can transform the first quantity so :

3.5 Cosmological Signatures of Inflation

Inflationary scenarios predict the existence of primordial perturbations generated by quantum fluctuations during the accelerated phase. These fluctuations stretch to cosmological scales with inflation, to become classical perturbations which correspond to anisotropies observed in the CMB in the distribution of matter and radiation.

Different inflationary potentials $U(\phi)$ lead to different results for observable quantities. The most important parameters are the **power spectrum amplitude A_s** , the **scalar spectral index n_s** , and the tensor-to-scalar ratio r . The scalar mode parameter n_s will correspond to the density fluctuations of the inflaton. Tensor mode fluctuations will be caused by primordial gravitational waves. It is interesting to notice that these tensor modes fluctuations are from the metric itself, not densities fluctuations. We can describe them with a perturbed metric $g_{ij} = a^2(t)(\delta_{ij} + h_{ij})$, with δ_{ij} the flat metric, and h_{ij} a small tensor perturbation which correspond to these gravitational waves. This describes small perturbations of spacetime.

Scalar mode perturbations are described by the comoving curvature perturbation $\mathcal{R}$. A comoving distance is a distance that will stay invariant with the expansion of space, as if it was stick to an expanding grid which gives a measure of distances. The power spectrum $P_s(k)$ give fluctuations quantities as a function of a comoving wavenumber k , which give a Fourier view on these perturbations in a frame that expand with the universe. Large k corresponds to small physical scales, while small k corresponds to large physical scales, because a fluctuation is a sum of $e^{i\vec{k}\cdot\vec{x}}$. At horizon crossing we have the calibration $k = aH$. In the slow-roll approximation, it reads

$$P_s(k) \simeq \frac{H^2}{8\pi^2 M_{\text{Pl}}^2 \epsilon} \simeq \frac{1}{24\pi^2 M_{\text{Pl}}^4} \frac{U}{\epsilon_U}. \quad (3.29)$$

A perfect scale-invariant spectrum would corresponds to $n_s = 1$, called "tilt" of the primordial fluctuation spectrum. Planck satellite observations[1] measured a spectral index of

$$n_s = 0.965 \quad (3.30)$$

meaning that the density fluctuations vary with scale, in this case more in big scales, thus the scale invariance is not perfect :

$$\begin{cases} n_s = 1 : \text{flat spectrum} \\ n_s < 1 : \text{downward tilt} \\ n_s > 1 : \text{upward tilt (not what we observe)} \end{cases} \quad (3.31)$$

Inflation may also produce primordial gravitational waves because of the perturbed metric as we saw before, which have amplitude measured with r . To notice the nature of primordial density perturbations we introduce an higher-order parameter called the running of the spectral index :

$$\alpha_s = \frac{dn_s}{d\ln(k)} \quad (3.32)$$

where k denotes the comoving Fourier mode associated with a perturbation scale. Tensor perturbations correspond to primordial gravitational waves with power spectrum

$$P_t(k) = \frac{2H^2}{\pi^2 M_{Pl}^2} \quad (3.33)$$

We define a tensor-to-scalar ratio which quantifies the relative amplitude of tensor perturbations compared to scalar perturbations :

$$r = \frac{P_t}{P_s} \quad (3.34)$$

With the slow-roll approximation we find the relation :

$$r = \frac{\frac{2H^2}{\pi^2 M_{Pl}^2}}{\frac{H^2}{8\pi^2 M_{Pl}^2 \epsilon}} \simeq 16\epsilon_U \quad (3.35)$$

This relation is very interesting because primordial gravitational waves are there directly correlated to the dynamics of the inflation potential.

The scalar spectral index is defined by

$$n_s - 1 = \frac{d \ln P_s}{d \ln k} \quad (3.36)$$

with slow-roll approximation it becomes :

$$n_s - 1 \simeq -6\epsilon_U + 2\eta_U \quad (3.37)$$

Cosmologists measure n_s because it constrains inflation models, inflation potential and give us information on the fluctuations of the very early universe, allowing inflationary models to be tested using CMB data. The deviation measured in the CMB is about :

$$1 - n_s \sim 0.035 \quad (3.38)$$

giving us information on the quantum fluctuations generated around $10^{-35}s$ after the Big Bang.

3.6 e-folds for quadratic potential

We use a quadratic potential :

$$U(\phi) = \frac{1}{2} m^2 \phi^2 \quad (3.39)$$

We calculate that for the end of inflation we have

$$\epsilon_f = \frac{2M_{Pl}^2}{\phi_f^2} = 1 \quad (3.40)$$

And so :

$$\phi_f = \sqrt{2}M_{Pl} \quad (3.41)$$

E-folds is a factor which describes the scale factor increase during inflation. One e-fold means that the scale factor increases by a factor e. We define it as :

$$N = \ln\left(\frac{a(t_f)}{a(t_i)}\right) \quad (3.42)$$

where $a(t_f)$ is the final scale factor and $a(t_i)$ the initial one.

$$\begin{aligned} N &= \frac{1}{M_{Pl}^2} \int_{\phi_f}^{\phi_i} \frac{U}{U'} d\phi = \frac{1}{M_{Pl}^2} \int_{\phi_f}^{\phi_i} \frac{\phi}{2} d\phi \\ &= \frac{1}{4M_{Pl}^2} (\phi_i^2 - \phi_f^2). \end{aligned} \quad (3.43)$$

We can inverse this equation and apply ϕ_f to get ϕ_i for $N \gg 1$:

$$\begin{aligned} \phi_i^2 &= 4NM_{Pl}^2 + \phi_f^2 = (4N + 2)M_{Pl}^2 \\ \phi_i &\simeq 2\sqrt{N} M_{Pl} \end{aligned} \quad (3.44)$$

3.7 Inflation models depending on n_s and r parameters

Different inflationary models can be compared according to their scalar spectral index, related to the tilt, and their tensor-to-scalar ratio. First we can find the expression for the slow-roll parameters at the horizon so ϕ_i and for N :

$$\epsilon_U = \frac{2M_{Pl}^2}{\phi_i^2} \approx \frac{1}{2N} \quad (3.45)$$

and we had $n_s - 1 \simeq -6\epsilon_U + 2\eta_U$ which means :

$$n_s \simeq 1 - \frac{2}{N} \quad r = \frac{8}{N} \quad (3.46)$$

This plot is very interesting because it gives us a direct link between theoretical description of the inflation and direct observations. The characteristics of the inflationary potential will directly shape the inflationary curve in the (r, n_s) diagram, depending also on the number of e-folds.

The diagram reveals that the quadratic chaotic model, have a relatively high tensor-to-scalar ratio ($r \simeq 8/N$), placing it under significant pressure from current CMB constraints. The Starobinsky model ($U(\phi) = \Lambda^4 \left(1 - e^{-\sqrt{2/3}\phi/M_{Pl}}\right)^2$) is aligning more with the observations on the spectral index $n_s \sim 0.965$, being localized in an higher probability region. The e-folds number N is typically localized between 50 and 60, changing model prediction along their trajectories in the plane. This allows theorists to directly align with data satellites, and confirm or compare their inflation models to observations.

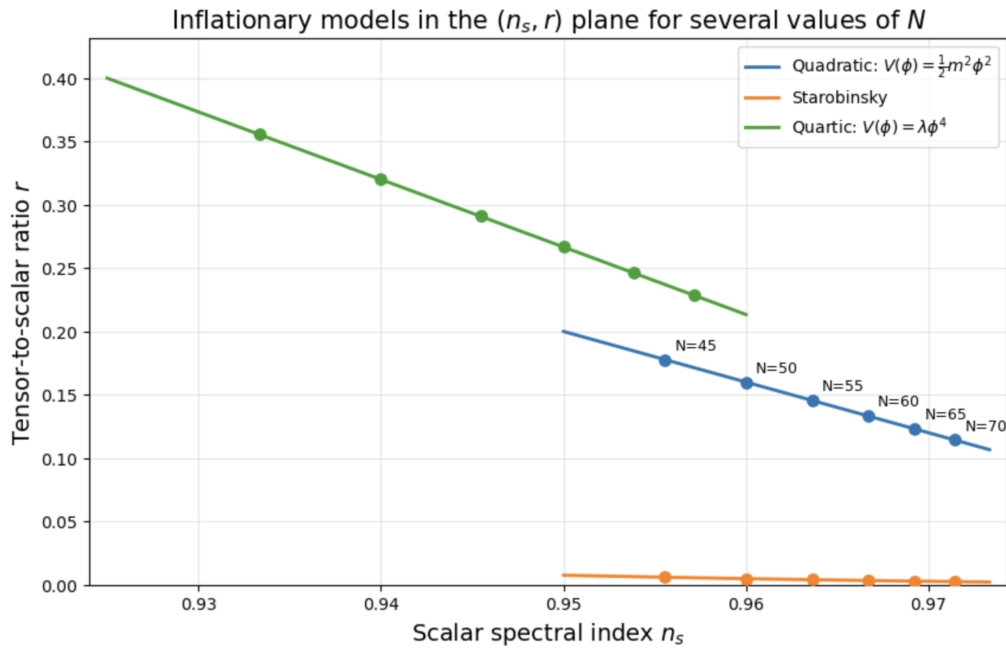


Figure 3.1: Different inflation potentials in the (r, n_s) diagram with different values of N .

Chapter 4

Conclusion

In this report we studied the tools necessary to study cosmic inflation and the way inflationary models are constrained by some cosmological observations. Introducing first the tools of the General Relativity of Einstein as the metric tensor, the Christoffel symbols, the geodesics, curvature tensors and the Einstein field equations. We applied these equations to a FLRW Universe to get the Friedmann equations, which describes the evolution of the scale factor.

After discussing the motivation behind inflation with the horizon and flatness problems, we introduce a scalar field with the inflatons. This field is a way to solve the problem of accelerated expansion with a potential energy which dominates over kinetic energy. In slow-roll regime, we saw that it links with simple observational constraints.

We finished by studying cosmological signatures of inflation with a scalar spectral index parameter n_s and the tensor-to-scalar ratio r . This allow us to compare different potentials with observations of the CMB. The quadratic chaotic model giving simple predictions, $n_s \simeq 1 - 2/N$ and $r \simeq 8/N$, was particularly constrained by current observational data.

This report shows how inflation connects high-energy physics, gravitational theory and precise cosmology. It is however important to specify that the exact nature of the inflaton field and its potential remains an open question. Future measurements of primordial gravitational waves and improved constraints on n_s , r , could be essential to choose or develop inflationary models.

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